SIMATS ENGINEERING

Engineer to Excel

SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO2 - ASSESSMENT 1

REPORT ANALYSIS

Course Faculty	**:**	**Dr. K.Anbazhagan**
Course Name	**:**	**Ethical Hacking**
Course Code	**:**	**ITA1405**
Slot	**:**	**Slot A**
Total Marks	**:**	**100**

CO2 :

Foot printing Tools: Conducting Competitive Intelligence, DNS Zone Transfers, Introduction to Social Engineering: Social Engineering Attacks and Countermeasures. (BT3)

Code of Conduct:

I M. SANA FATHIMA(192371082) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student
M SANA FATHIMA

Answer the Following

Kindly follow the below step to do Domain & Email Footprinting using theHarvester

Step 1: Install Python

- Open https://www.python.org/downloads/
- Download the latest Python version.
- Run the installer.
- Tick Add Python to PATH • Click Install Now

Step 2: Verify Python Installation

- Press Windows + R
- Type cmd and press Enter
- Run: [python -m pip install --upgrade pip] in cmd

Step 4: Install theHarvester

- Run: [pip install theHarvester] in cmd
- Run: [python -m pip install theHarvester] in cmd
- Wait until the installation Tinishes.

Step 5: Verify theHarvester

- Run: [theHarvester --help] in cmd

Step 6: Run theHarvester

- Run: [theHarvester -d saveetha.com -b all] in cmd

Step 7: Wait for the Results

- After a few minutes, theHarvester displays information such as:
- Email Addresses
- Subdomains
- Host Names
- IP Addresses

TOOLS AND TECHNOLOGIES USED

The following tools and technologies were used:

- **Python:** Provides the Python environment required by the project.
- **uv:** Manages the project's Python environment and dependencies.
- **theHarvester:** Performs domain and email footprinting using supported data sources.
- **Windows Command Prompt:** Used to execute commands and inspect tool output.
- **Internet Connection:** Required to access public information sources.

5. TARGET DOMAIN

Domain Name: saveetha.com

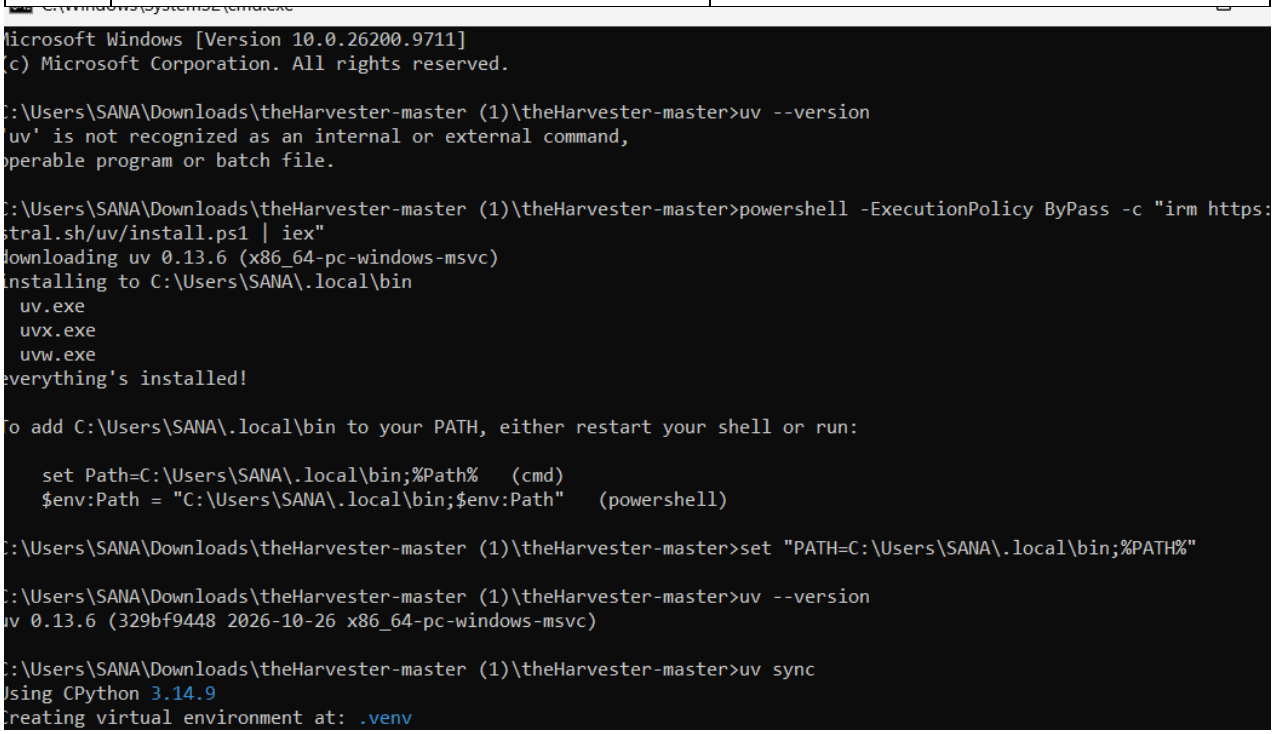
The domain specified in the assessment was used as the target for the footprinting experiment.

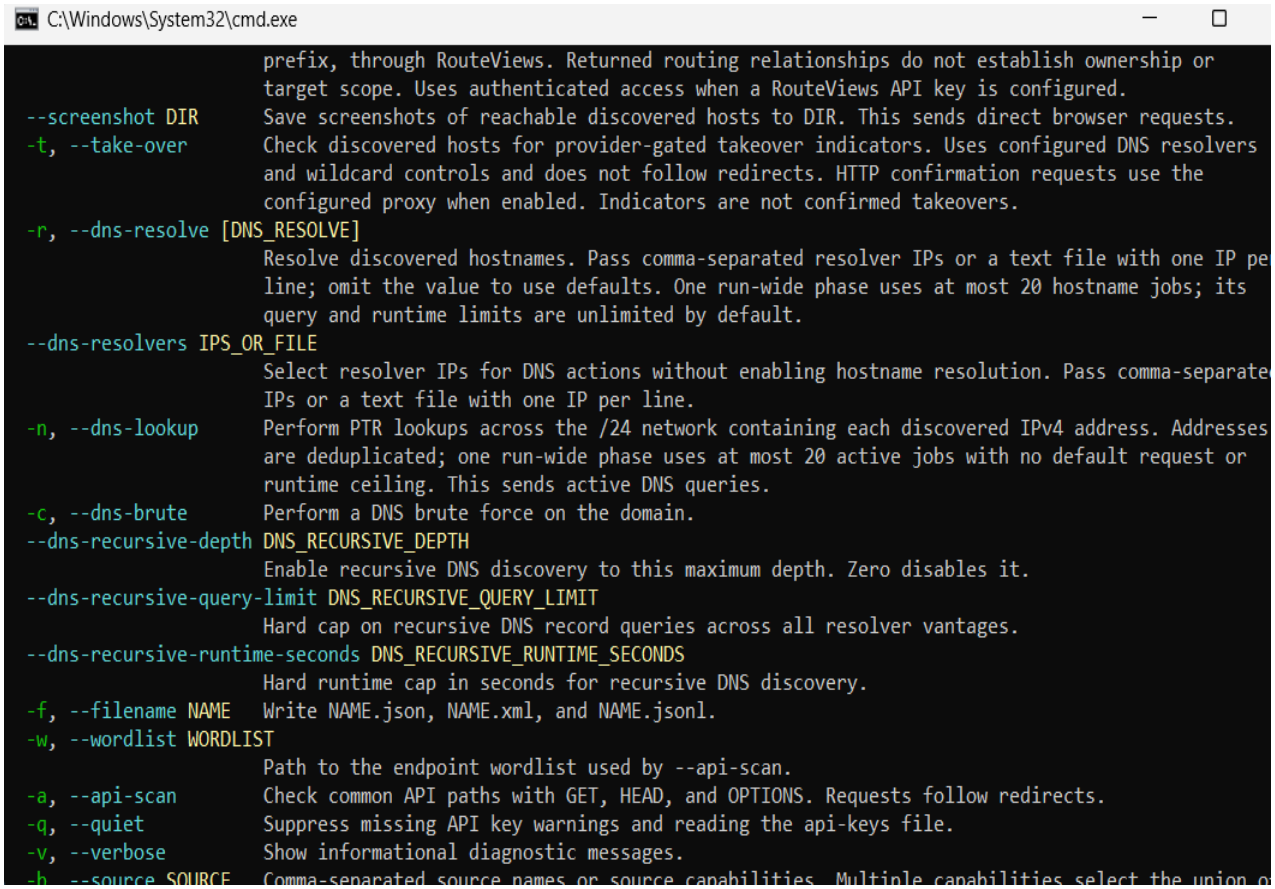
The tool was executed to collect information from its available data sources. The results documented in this report are limited to the output obtained during the experiment.

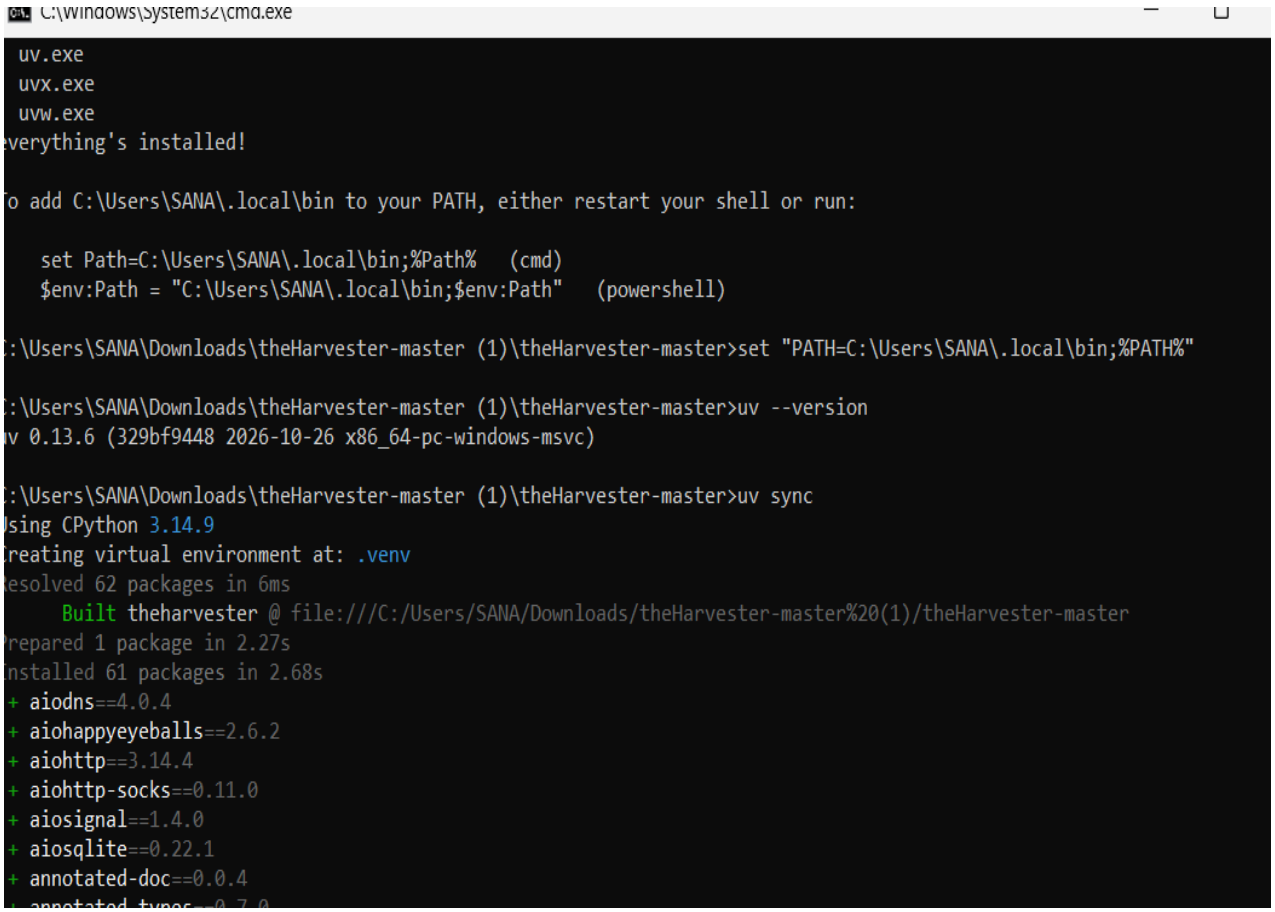
COMMANDS EXECUTED

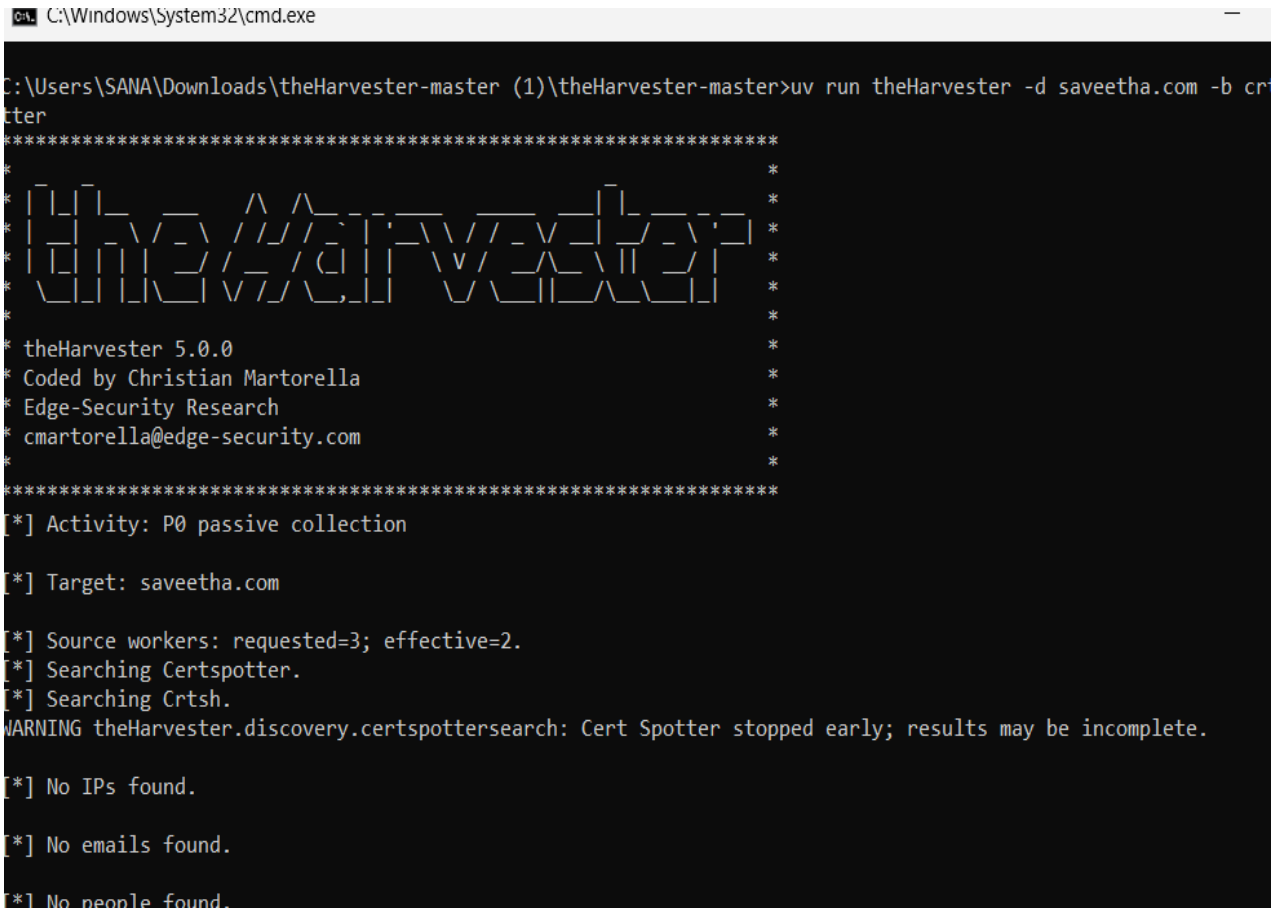
The following commands were used during the experiment.

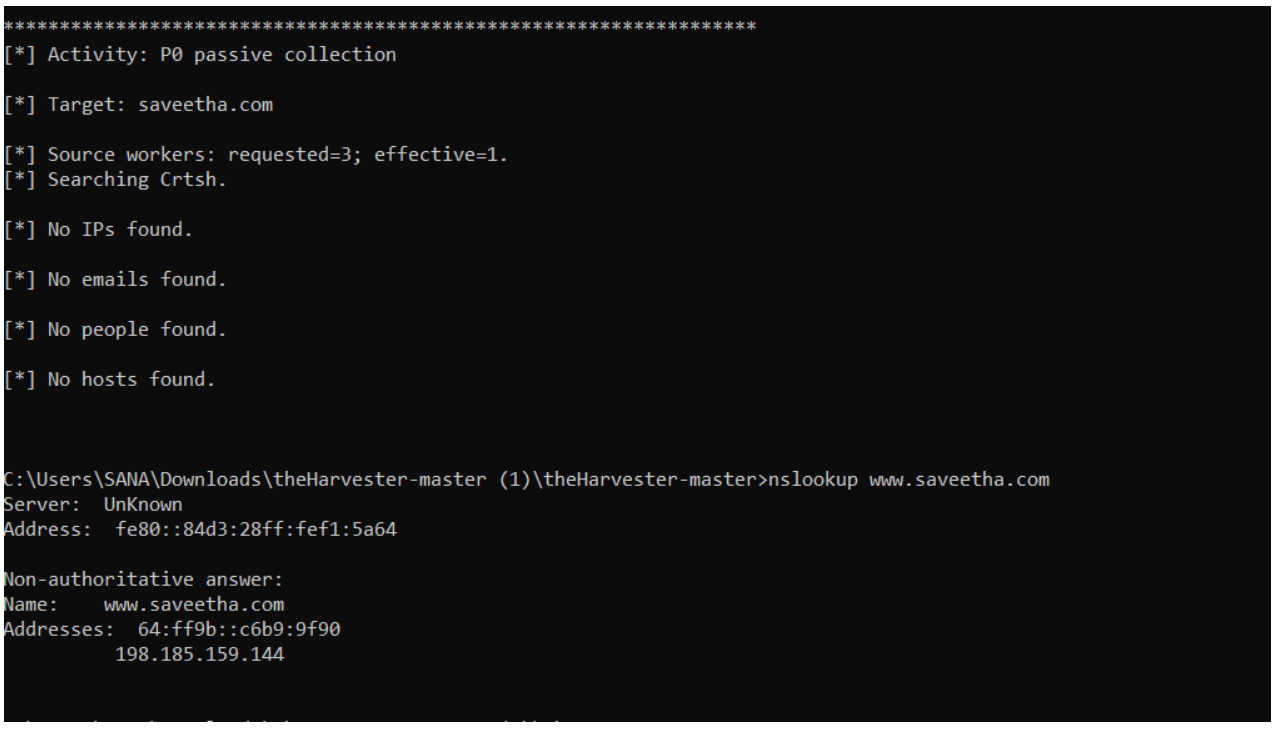
S. No.	Command	Purpose
1	python --version	Verify Python installation
2	python -m pip --version	Verify pip availability
3	python -m pip install --upgrade pip	Check and upgrade pip
4	uv --version	Verify uv installation
5	uv sync	Install project dependencies
6	uv run theHarvester -h	Display tool help information
7	uv run theHarvester -d saveetha.com -b all	Perform domain and email footprinting




C:\Windows\System32\cmd.exe
prefix, through RouteViews. Returned routing relationships do not establish ownership or target scope. Uses authenticated access when a RouteViews API key is configured.
--screenshot DIR Save screenshots of reachable discovered hosts to DIR. This sends direct browser requests.
-t, --take-over Check discovered hosts for provider-gated takeover indicators. Uses configured DNS resolvers and wildcard controls and does not follow redirects. HTTP confirmation requests use the configured proxy when enabled. Indicators are not confirmed takeovers.
-r, --dns-resolve [DNS_RESOLVE]
Resolve discovered hostnames. Pass comma-separated resolver IPs or a text file with one IP pe line; omit the value to use defaults. One run-wide phase uses at most 20 hostname jobs; its query and runtime limits are unlimited by default.
--dns-resolvers IPS_OR_FILE
Select resolver IPs for DNS actions without enabling hostname resolution. Pass comma-separate IPs or a text file with one IP per line.
-n, --dns-lookup Perform PTR lookups across the /24 network containing each discovered IPv4 address. Addresses are deduplicated; one run-wide phase uses at most 20 active jobs with no default request or runtime ceiling. This sends active DNS queries.
-c, --dns-brute Perform a DNS brute force on the domain.
--dns-recursive-depth DNS_RECURSIVE_DEPTH
Enable recursive DNS discovery to this maximum depth. Zero disables it.
--dns-recursive-query-limit DNS_RECURSIVE_QUERY_LIMIT
Hard cap on recursive DNS record queries across all resolver vantages.
--dns-recursive-runtime-seconds DNS_RECURSIVE_RUNTIME_SECONDS
Hard runtime cap in seconds for recursive DNS discovery.
-f, --filename NAME Write NAME.json, NAME.xml, and NAME.jsonl.
-w, --wordlist WORDLIST
Path to the endpoint wordlist used by --api-scan.
-a, --api-scan Check common API paths with GET, HEAD, and OPTIONS. Requests follow redirects.
-q, --quiet Suppress missing API key warnings and reading the api-keys file.
-v, --verbose Show informational diagnostic messages.


C:\Windows\System32\cmd.exe
uv.exe
uvx.exe
uvw.exe
everything's installed!
To add C:\Users\SANA\.local\bin to your PATH, either restart your shell or run:
set Path=C:\Users\SANA\.local\bin;%Path% (cmd)
$env:Path = "C:\Users\SANA\.local\bin;$env:Path" (powershell)
C:\Users\SANA\Downloads\theHarvester-master (1)\theHarvester-master>set "PATH=C:\Users\SANA\.local\bin;%PATH%"
C:\Users\SANA\Downloads\theHarvester-master (1)\theHarvester-master>uv --version
uv 0.13.6 (329bf9448 2026-10-26 x86_64-pc-windows-msvc)
C:\Users\SANA\Downloads\theHarvester-master (1)\theHarvester-master>uv sync
Using CPython 3.14.9
Creating virtual environment at: .venv
Resolved 62 packages in 6ms
Built theharvester @ file:///C:/Users/SANA/Downloads/theHarvester-master%20(1)/theHarvester-master
Prepared 1 package in 2.27s
Installed 61 packages in 2.68s
+ aiodns==4.0.4
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.14.4
+ aiohttp-socks==0.11.0
+ aiosignal==1.4.0
+ aiosqlite==0.22.1
+ annotated-doc==0.0.4


C:\Windows\System32\cmd.exe
C:\Users\SANA\Downloads\theHarvester-master (1)\theHarvester-master>uv run theHarvester -d saveetha.com -b cr
tter
theHarvester 5.0.0
Coded by Christian Martorella
Edge-Security Research
cmartorella@edge-security.com
[*] Activity: P0 passive collection
[*] Target: saveetha.com
[*] Source workers: requested=3; effective=2.
[*] Searching Certspotter.
[*] Searching Crtsh.
WARNING theHarvester.discovery.certspottersearch: Cert Spotter stopped early; results may be incomplete.
[*] No IPs found.
[*] No emails found.
[*] No people found.


[*] Activity: P0 passive collection
[*] Target: saveetha.com
[*] Source workers: requested=3; effective=1.
[*] Searching Crtsh.
[*] No IPs found.
[*] No emails found.
[*] No people found.
[*] No hosts found.
C:\Users\SANA\Downloads\theHarvester-master (1)\theHarvester-master>nslookup www.saveetha.com
Server: UnKnown
Address: fe80::84d3:28ff:fef1:5a64
Non-authoritative answer:
Name: www.saveetha.com
Addresses: 64:ff9b::c6b9:9f90
198.185.159.144

FOOTPRINTING RESULTS TABLE

The following table records the results supported by the captured output.

S. No.	Required Detail	Result
1	Domain Name	saveetha.com
2	Emails Found	No emails found in the captured output
3	Sub Emails Found	No email addresses returned in the captured output
4	Hosts and Their IP Addresses	No hosts found in the captured output
5	Total Email Addresses Found	0 in the captured output
6	Total Subdomains Found	To be verified from the complete output
7	Total Hosts Found	0 in the captured output
8	Total IP Addresses Found	0 in the captured output

RUBRICS

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (ProYicient)	Level 2 (Developing)	Level 1 (Beginning)
Tool Installation and ConYiguration	25%	Successfully installs, veriTies, and conTigures theHarvester and required environment without errors.	Installs and conTigures the tool correctly with minor issues.	Completes installation with assistance and encounters several conTiguration issues.	Unable to install, conTigure, or verify the tool.
Footprinting Data Collection	25%	Accurately collects and organizes emails, sub-emails, hosts, IP addresses, and subdomains comprehensively.	Collects most required footprinting information with minor omissions.	Collects limited information or misses several required Tields.	Fails to obtain meaningful footprinting information.
Analysis and Interpretation of Results	30%	Thoroughly analyzes theHarvester output and accurately interprets the signiTicance of discovered information.	Correctly interprets most results with minor errors.	Provides basic interpretation with limited technical understanding.	Unable to interpret the collected footprinting information.

Evidence and Documentation	20%	Provides complete screenshots, counts, lists, and organized documentation with technically accurate presentation.	Provides adequate evidence and documentation with minor omissions.	Documentation is partially complete or poorly organized.	Evidence is missing, incomplete, or difficult to verify.